Patuakhali Science and Technology University
Faculty of Computer Science and Engineering

CIT 320 :: Software Development Project-II

Project Proposal

Project Title : Reuse Hub
Submission Date : Thu, 27 Jan 2026

Submitted from,	Submitted to,
Prosenjit Mondol **ID** : 2102049, **Reg** : 10176, **Semester** : 6 (Level-3, Semester-2)	1. **Md Mahbubur Rahman** Associate Professor, Department of Computer Science and Information Technology, Patuakhali Science and Technology University. 2. **Md Atikqur Rahaman** Professor, Department of Computer Science and Information Technology, Patuakhali Science and Technology University.

Contents

ReuseHub

1. Abstract

Reuse Hub is a cross-platform application designed to promote sustainable living through community-driven item reuse. It connects donors and seekers, enabling free exchange of usable goods with an intuitive and secure interface [1]. Built using Flutter and powered by Supabase, the app ensures real-time interactions and platform independence. Key features include item listings, request handling, messaging, and role-based access [2]. Reuse Hub aims to reduce waste while fostering a culture of sharing and environmental responsibility [3], [4].

2. Objectives

- To develop a unified, cross-platform platform for donating and requesting reusable household items [1].
- Implement secure login and user roles.
- Enable item listing, browsing, and filtering.
- Support direct requests and real-time messaging.
- Ensure responsive design across mobile and web.

3. Job Market Analysis

The job market for mobile application developers in Bangladesh is growing rapidly, with a surge in demand for cross-platform technologies like Flutter. Companies are increasingly adopting Flutter to reduce development costs by maintaining a single codebase for both iOS and Android. According to major job portals:

Job Portal	**Link**	**Stack**
BD Jobs	• https://bdjobs.com/jobs/details/flutter-developer • https://bdjobs.com/jobs/details/mobile-app-developer	Flutter, Dart, Firebase, REST API, Git
LinkedIn (Bangladesh)	• https://www.linkedin.com/jobs/view/junior-flutter-engineer • https://www.linkedin.com/jobs/view/mobile-engineer-dhaka	Flutter, GetX/Bloc, Supabase, Figma
Glassdoor & Wellfound	• https://wellfound.com/jobs/remote-flutter-dev • https://www.glassdoor.com/Job/bangladesh-flutter-jobs	Flutter, CI/CD, Unit Testing, Riverpod

Table 1: Job Market Opportunities for Flutter & Mobile App Developers in Bangladesh

4. Related Work

- **OLIO**: A widely-used food and item sharing app.
- **Freecycle**: A nonprofit global network for item reuse, but suffers from outdated UI, limited mobile experience, and lacks in-app messaging.
- **Letgo (Now OLX)**: Supported local item sharing but shifted focus to resale, reducing emphasis on community reuse [5].
- **Trash Nothing**: Focused purely on free giving, but lacks a modern UI and role-based access.
- **Facebook Marketplace**: Highly popular but lacks filters specifically for “free” items.
- **Donate Kart**: Focuses on donating essentials but operates differently from a direct peer-to-peer exchange.
- **Too Good to Go**: Focuses only on surplus food donations, with no support for general household items.

5. Scope

Reuse Hub is a cross-platform application built using Flutter, designed to facilitate the reuse and redistribution of pre-owned items within communities. The platform will enable users to donate or request items through a streamlined, user-friendly interface. Core functionalities include user authentication, item listings, request handling, messaging, and category-based browsing.

6. Methodology

6.1. Technology Stack

The development of Reuse Hub will follow an agile methodology, allowing for iterative improvements and user feedback. Our technology stack will include:

- **Frontend**: Flutter (Dart)
- **Backend**: Supabase (Alternative to Firebase)
- **UI/UX Design**: Figma
- **Database**: PostgreSQL
- **Hosting (web app)**: Vercel or Netlify
- **CI/CD**: GitHub Actions, Docker (optional)

6.2. Design Principles

The design of Reuse Hub will adhere to the following principles:

- **Material Design**: Adopts Google’s Material Design guidelines to ensure a consistent and modern UI/UX across Android devices.
- **Responsive Design**: Guarantees seamless performance across various screen sizes and orientations, including desktop, tablet, and mobile views [1].
- **User-Centric Design**: Prioritizes user experience with intuitive navigation, minimal steps, and clear visual elements for both donors and receivers.
- **Cross-Platform**: Ensures consistent design and functionality across Android, iOS, and web platforms using a unified development approach.
- **Documentation**: Provides thorough documentation for developers (e.g., API docs) and end-users (e.g., help guides, FAQs).

7. Visual Models

7.1. Flow Chart Diagram

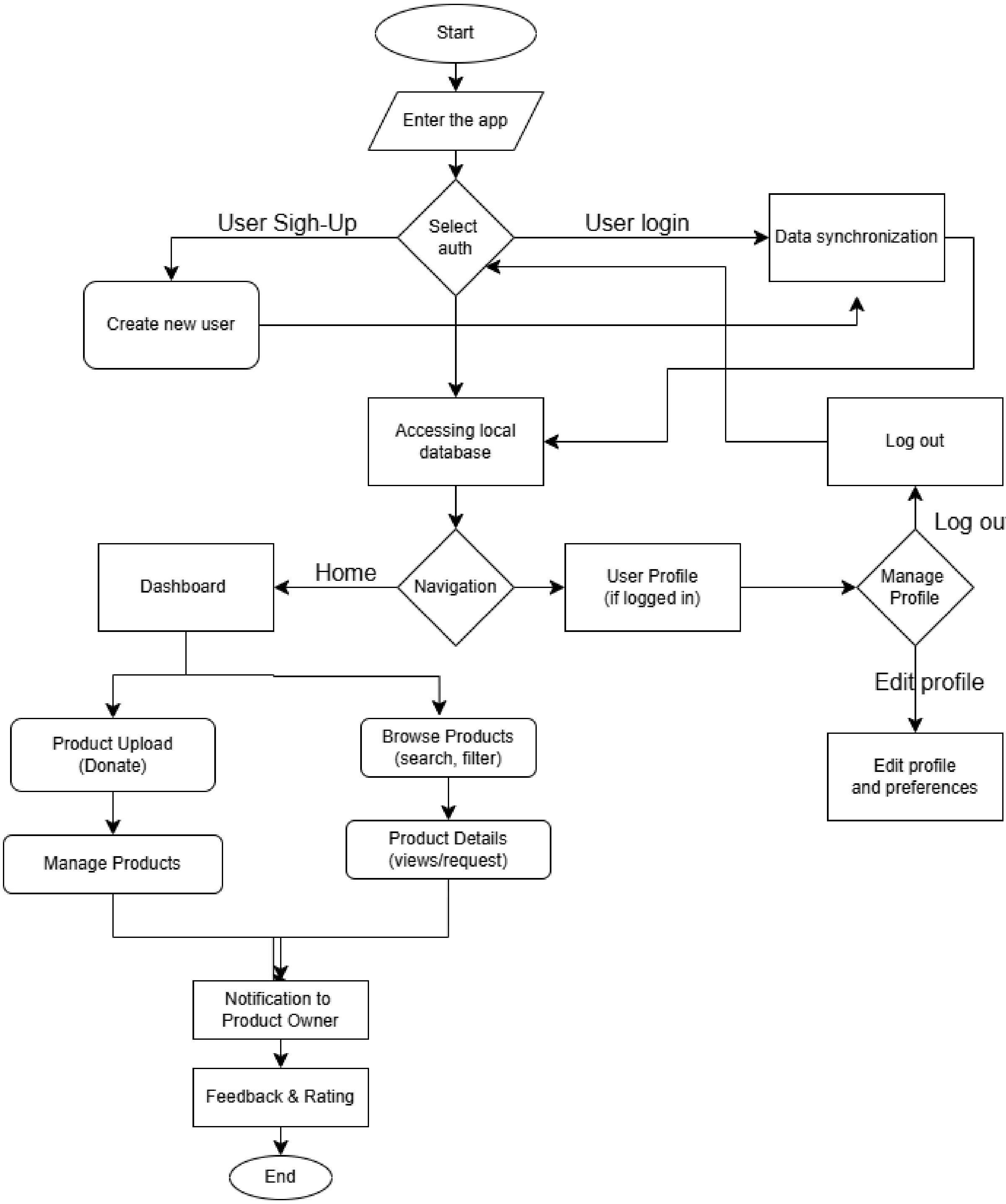

Figure 1: Flow Chart of Reuse Hub Architecture

Figure 1 illustrates the high-level user interaction flow of Reuse Hub, showcasing how various components such as item listing, request handling, and user roles (Donor and Seeker) operate within the app. The diagram primarily focuses on the frontend-driven flow, from authentication to donation completion.

7.2. ERD (Entity Relationship Diagram)

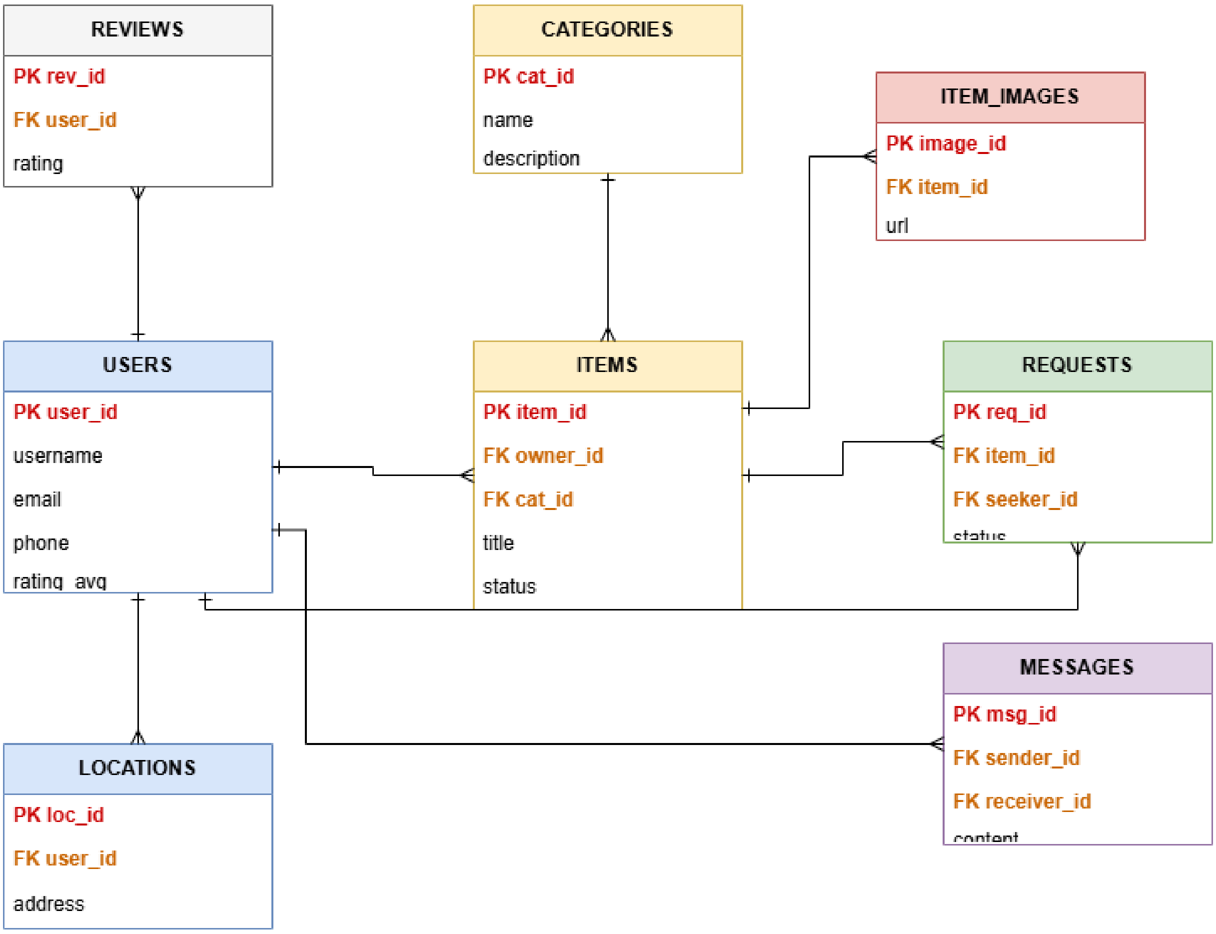

Figure 2: Entity Relationship Diagram of Reuse Hub

The Figure 2 illustrates the database schema for Reuse Hub, showing the tables, their fields, and relationships between them.

7.3. Timeline (Gantt Chart)

The base timeline for the development of Reuse Hub is as follows:

Task	Wk 1-4	Wk 5-7	Wk 8-9	Wk 10-11	Wk 12	Wk 14	Wk 15	Wk 16
Requirements & UI Wireframing	✓		✓					
Authentication + DB		✓	✓					
Item Listing & Upload Module			✓	✓				
Nearby Item Discovery				✓				
Request & Donation Management			✓	✓	✓			
Chat & Notification System						✓		
UI Polish + Accessibility						✓	✓	
Final Testing & Deployment								✓

Table 2: Development Timeline of Reuse Hub

7.4. UI Mockups

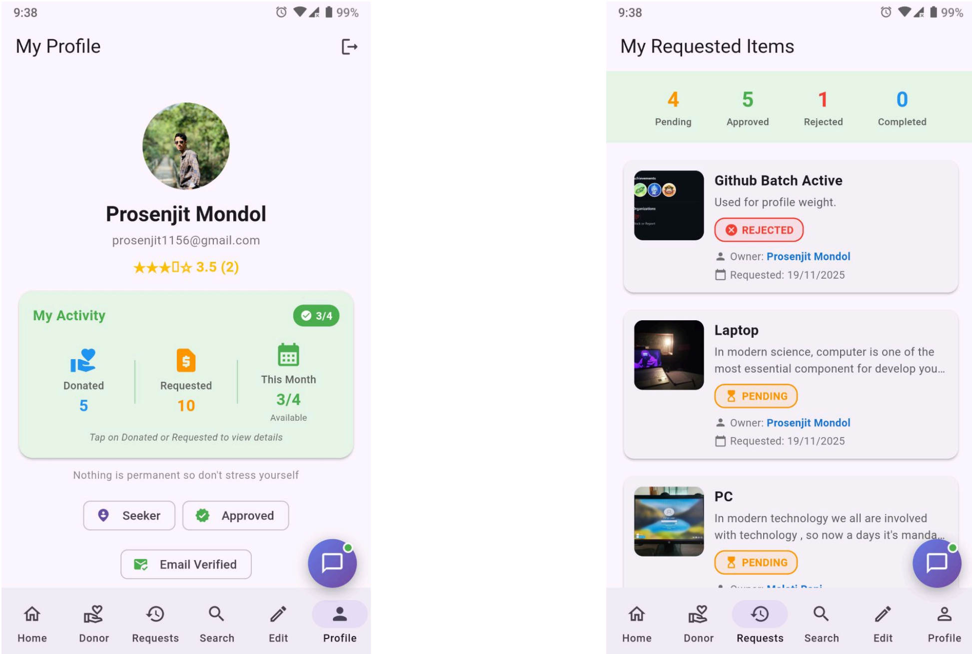

Figure 3: Login and Sign-Up page UI concept

8. Limitations

1. User base growth is dependent on community awareness and participation, which may be slow initially [2].
2. The app relies on accurate location data; privacy concerns or GPS inaccuracies may affect user experience.
3. Lack of a formal verification system for item quality or donor credibility.

9. Result

The expected outcome of Reuse Hub is a fully functional, cross-platform application that enables users to donate and collect reusable household items efficiently.

10. References

[1] J. Smith, "Designing Cross-Platform Mobile Apps with Flutter," *International Journal of Software Engineering*, vol. 12, no. 3, pp. 45–56, Mar. 2024.

[2] L. Chen and M. Gupta, "Community-Driven Sustainability Initiatives: A Case Study," *Journal of Environmental Studies*, vol. 10, no. 2, pp. 120–130, Feb. 2023.

[3] United Nations, "Sustainable Consumption and Production." [Online]. Available: https://www.un.org/sustainabledevelopment/sustainable-consumption-production/

[4] Ellen MacArthur Foundation, "What is a Circular Economy?." [Online]. Available: https://ellenmacarthurfoundation.org/topics/circular-economy-introduction/overview

[5] OLX, "Online Marketplace Bangladesh." [Online]. Available: https://www.olx.com/